

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

**Time Duration: 1 Hour
Total Marks:30**

QUESTIONS: 5

**Annexure A
Resolution Algorithm Implementation**

Code of Conduct:

*I **Anchal Kumari(192312255)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Anchal Kumari

Annexure A

Resolution Algorithm Implementation

Question 1 — Rain and Wet Ground

1. Propositional Symbols

- R = It is raining
- W = The ground is wet

2. Knowledge Base (English $\rightarrow$ Logic)

1. If it rains, then the ground becomes wet. $\rightarrow R \rightarrow W$
2. It is raining. $\rightarrow R$

3. Conversion to CNF

- Clause 1: $\neg R \vee W$
- Clause 2: R

4. Negate the Goal

Negated Goal: $\neg W$ (added as Clause 3)

5. Resolution Steps

Step	Clauses Used	Resolvent
1	$\neg R \vee W$ (C1) and $\neg W$ (C3)	$\neg R$
2	$\neg R$ (from Step 1) and R (C2)	$\square$ (Empty Clause)

6. Conclusion

Since resolution produced the Empty Clause ($\square$), the negated goal $\neg W$ leads to a contradiction with the KB. Therefore the original goal 'The ground is wet (W)' is PROVED.

Question 2 — Student Assignment Submission

1. Propositional Symbols

- S = Rahul submits the assignment
- M = Rahul receives internal marks

2. Knowledge Base (English $\rightarrow$ Logic)

1. If a student submits the assignment, then the student receives internal marks. $\rightarrow S \rightarrow M$
2. Rahul submitted the assignment. $\rightarrow S$

3. Conversion to CNF

- Clause 1: $\neg S \vee M$
- Clause 2: S

4. Negate the Goal

Negated Goal: $\neg M$ (added as Clause 3)

5. Resolution Steps

Step	Clauses Used	Resolvent
1	$\neg S \vee M$ (C1) and $\neg M$ (C3)	$\neg S$
2	$\neg S$ (from Step 1) and S (C2)	$\square$ (Empty Clause)

6. Conclusion

The Empty Clause ($\square$) is derived, so $\neg M$ contradicts the KB. Therefore 'Rahul receives internal marks (M)' is PROVED.

Question 3 — Library Membership

1. Propositional Symbols

- L = Priya is a library member
- B = Priya can borrow books

2. Knowledge Base (English $\rightarrow$ Logic)

1. If a person is a library member, then the person can borrow books. $\rightarrow L \rightarrow B$
2. Priya is a library member. $\rightarrow L$

3. Conversion to CNF

- Clause 1: $\neg L \vee B$
- Clause 2: L

4. Negate the Goal

Negated Goal: $\neg B$ (added as Clause 3)

5. Resolution Steps

Step	Clauses Used	Resolvent
1	$\neg L \vee B$ (C1) and $\neg B$ (C3)	$\neg L$
2	$\neg L$ (from Step 1) and L (C2)	$\square$ (Empty Clause)

6. Conclusion

No further clauses can be generated after reaching $\square$. Since the negated goal leads to contradiction, 'Priya can borrow books (B)' is PROVED.

Question 4 — Placement Eligibility

1. Propositional Symbols

- A = Arun clears the aptitude test
- E = Arun is eligible for placement

2. Knowledge Base (English $\rightarrow$ Logic)

1. If a student clears the aptitude test, then the student is eligible for placement. $\rightarrow A \rightarrow E$
2. Arun cleared the aptitude test. $\rightarrow A$

3. Conversion to CNF

- Clause 1: $\neg A \vee E$
- Clause 2: A

4. Negate the Goal

Negated Goal: $\neg E$ (added as Clause 3)

5. Resolution Steps

Step	Clauses Used	Resolvent
1	$\neg A \vee E$ (C1) and $\neg E$ (C3)	$\neg A$
2	$\neg A$ (from Step 1) and A (C2)	$\square$ (Empty Clause)

6. Conclusion

Resolution Tree: $C1(\neg A \vee E) + C3(\neg E) \rightarrow \neg A$; $\neg A + C2(A) \rightarrow \square$. The Empty Clause proves the negated goal false, so 'Arun is eligible for placement (E)' is PROVED.

Question 5 — Access Control System

1. Propositional Symbols

- P = User enters the correct password
- Q = User is authenticated
- R = User is granted access

2. Knowledge Base (English $\rightarrow$ Logic)

1. If a user enters the correct password, then the user is authenticated. $\rightarrow P \rightarrow Q$
2. If a user is authenticated, then the user is granted access. $\rightarrow Q \rightarrow R$
3. The user entered the correct password. $\rightarrow P$

3. Conversion to CNF

- Clause 1: $\neg P \vee Q$
- Clause 2: $\neg Q \vee R$
- Clause 3: P

4. Negate the Goal

Negated Goal: $\neg R$ (added as Clause 4)

5. Resolution Steps

Step	Clauses Used	Resolvent
1	$\neg Q \vee R$ (C2) and $\neg R$ (C4)	$\neg Q$
2	$\neg P \vee Q$ (C1) and $\neg Q$ (from Step 1)	$\neg P$
3	$\neg P$ (from Step 2) and P (C3)	$\square$ (Empty Clause)

6. Conclusion

Each step resolves one literal pair at a time: first eliminating R , then Q , then P , until the Empty Clause is reached. This confirms the negated goal is unsatisfiable, so 'The user is granted access (R)' is PROVED.